Software Test Plan (STP) - Online Internship & Placement Preparation Portal

Project: Online Internship & Placement Preparation Portal (OIPPP)

Version: 1.0

Authors: Deekshith Gowda G, Sharath H G, Guru R Bharadwaj, E S Sukruth

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1. Introduction

Purpose: This plan describes how we will test OIPPP release 1.0: what we test, how, with which tools, on what schedule and who does it.

Scope: Testing covers the web client, the REST API, the MongoDB data rules and the analytics service for all eight features in the SRS, plus the performance, safety and security requirements. The SMTP provider and MongoDB Atlas themselves are not tested.

References: OIPPP SRS v1.0, OIPPP SAD v1.0, OWASP Top 10 (2021), OWASP ASVS 4.0, IEEE 829-2008.

Definitions: SRS (Software Requirements Specification), SAD (Software Architecture and Design), RTM (Requirements Traceability Matrix), TC (test case), UAT (user acceptance test), JWT (JSON Web Token).

2. Test Items

- Auth and user module (registration, login, lockout, profiles, resume upload, recruiter approval)
- Listing module (CRUD, deadline closing job, search, filters, bookmarks, feed)
- Application module (apply, status state machine, applicant view) and notification module
- Aptitude test module (question bank, test builder, attempts, auto-save, scoring)
- Analytics gateway and Python analytics service
- React client screens for the three roles
- Build under test: the commit tagged for each sprint review on the dev branch

3. Features to be Tested

Features mapped to SRS requirement IDs:

SRS feature	Requirement IDs	Priority	Test owner
4.1 Authentication and Role-Based Access	AUTH-1, AUTH-2, AUTH-3, AUTH-4, AUTH-5, AUTH-6	High	Deekshith Gowda G
4.2 Profile and User Administration	PROF-1, PROF-2, PROF-3, PROF-4	High	Deekshith Gowda G
4.3 Internship Listing Management	LST-1, LST-2, LST-3, LST-4	High	Sharath H G
4.4 Search, Filtering, Bookmarks and Recommendations	SRCH-1, SRCH-2, SRCH-3, SRCH-4, SRCH-5	High	Sharath H G
4.5 Application Submission and Tracking	APP-1, APP-2, APP-3, APP-4, APP-5	High	Guru R Bharadwaj
4.6 Recruiter Review and Notifications	REV-1, REV-2, REV-3, REV-4	Medium	Guru R Bharadwaj

SRS feature	Requirement IDs	Priority	Test owner
4.7 Aptitude Test Engine	TST-1, TST-2, TST-3, TST-4, TST-5, TST-6	High	E S Sukruth
4.8 Progress Tracking and Analytics	ANL-1, ANL-2, ANL-3	Medium	E S Sukruth
5.1 Performance	PERF-1, PERF-2, PERF-3	High	Sharath H G
5.2 Safety	SAFE-1, SAFE-2	High	E S Sukruth
5.3 Security	SEC-1 to SEC-6	High	Deekshith Gowda G
5.4 Quality attributes	QA-1, QA-3, QA-4, QA-5	Medium	All

4. Features Not to be Tested

- MongoDB Atlas backups and availability (provider's responsibility).
- Delivery inside the SMTP provider after it accepts the mail; we test only that the mail is handed over (checked in a Mailtrap sandbox).
- QA-2 availability over a full placement season; it can't be measured within the semester. We only run the uptime monitor during the demo weeks.
- Browsers other than the latest two versions of Chrome, Firefox and Edge.
- Load beyond 50 concurrent users; the free tier cannot handle it and it is beyond what the SRS asks for.

5. Test Approach / Strategy

Levels:

- Unit tests: each service function with Jest, database replaced by mongodb-memory-server. Written by the module owner in the same sprint as the code.
- Integration tests: API routes with Supertest against an in-memory database, including middleware (auth, validation, errors). Run in GitHub Actions on every pull request.
- System tests: the deployed test build exercised end to end through the UI and Postman, using the test cases in section 13.
- Acceptance tests: in Sprint 3 five classmates act as students and one as a recruiter, following task scripts (QA-1).

Types:

- Functional testing of every REQ in section 3, with positive, negative and boundary cases.
- Regression testing: the full Jest suite runs on every PR; the Postman collection is rerun before each sprint review.
- Performance testing with JMeter against 5,000 seeded listings (PERF-1, PERF-2).
- Usability testing with timed task scripts and a short questionnaire.
- Security testing, described in 5.1.

Every PR needs green CI and one reviewer. A defect found in system testing gets a Jira Bug linked to the story and the failed TC.

Entry criteria: Stable build deployed to the test environment, seed data loaded, test cases for the sprint's stories reviewed, Jira stories in 'Ready for QA'.

Exit criteria: All planned TCs for the sprint executed, no open Critical or High bugs, statement coverage on the server at least 70% (QA-4), every REQ in the RTM has at least one passed test.

5.1 Security Validation

- Password storage: read stored user documents and confirm bcrypt hashes with cost 10 or more; grep logs for passwords and tokens (SEC-1).
- Transport: check that HTTP redirects to HTTPS and that only TLS 1.2+ is offered, using SSL Labs on the deployed URL (SEC-2).
- Access control matrix: call every route with each role and with no token; anything not in the SAD API table must return 401 or 403. Includes resume access and listing ownership (AUTH-6, SEC-3).
- Injection: send $ne, $gt and $where operators in JSON bodies and query strings; send script tags in text fields and confirm they are rendered as text (SEC-4).
- Brute force: script 11 login attempts per minute and 6 wrong passwords in a row; expect 429 and 423 (AUTH-5, SEC-5).
- Audit: change statuses and admin actions, then confirm the audit entries exist and no API route edits or deletes them (SEC-6).
- Automated scan: OWASP ZAP baseline scan on the test build every sprint; High findings become Jira bugs.
- Dependencies: npm audit and pip-audit in CI; builds fail on High or Critical advisories.

6. Test Environment

Hardware: Developer laptops (8 GB RAM, Windows 11 / Ubuntu 22.04) for unit and integration tests; Render free instance and Atlas M0 for system tests; one 4 vCPU VM for JMeter.

Software: Node.js 20, MongoDB 7, Python 3.11, latest Chrome, Firefox and Edge, Android Chrome on a 360 px phone.

Tools: Jest, Supertest, mongodb-memory-server, Postman/Newman, JMeter 5.6, OWASP ZAP, Mailtrap, GitHub Actions, Jira (test execution and defects).

Test Data: Seed script creating 1 admin, 10 recruiters, 200 students with SRNs and CGPAs, 5,000 listings, 100 questions and 5 tests. No real student data is used.

7. Test Schedule

Milestones (aligned with the Jira sprints):

Milestone	Date
Test plan and first test cases	07-Oct-2026
Sprint 1 (auth, listings CRUD) testing	12-Oct-2026 to 18-Oct-2026
Sprint 2 (search, applications, tests) testing	26-Oct-2026 to 01-Nov-2026
Sprint 3 (analytics, notifications) testing, performance and ZAP scan	09-Nov-2026 to 15-Nov-2026
System test execution end	18-Nov-2026
UAT	19-Nov-2026 to 21-Nov-2026
Test summary report	23-Nov-2026

8. Test Deliverables

- This test plan
- Test cases (section 13 and Jira)
- Jest and Supertest suites in server/tests
- Postman collection and JMeter plan
- Seed data script

- CI run logs and coverage reports
- Defect reports in Jira
- Test summary report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Guru R Bharadwaj	Owns this plan, RTM and CI test pipeline; coordinates system test runs
Test Engineer (security)	Deekshith Gowda G	Security validation in 5.1, access control matrix
Test Engineer (performance)	Sharath H G	Seed data, JMeter plan, performance runs
Test Engineer (reliability, UAT)	E S Sukruth	Test attempt reliability cases, UAT sessions
Developer (all)	All members	Unit and integration tests for own modules, fix and retest own defects
Product Owner	Course faculty / placement cell contact	Accept sprint demos and UAT results

10. Risks and Mitigation

Risk	Mitigation
Features land late in the sprint, leaving no time to test	Stories are not 'Done' in Jira without passing tests; test cases written at sprint planning
Free hosting sleeps or is slow during test runs	Run system tests against a local Docker setup when Render is down; record which environment was used
Performance results on free tier are not representative	Run JMeter on the VM with the same build and note the difference
Team members busy with other course work	Each module has a backup tester named in Jira
Flaky timer-based tests (auto-close, auto-submit)	Inject a fake clock in services instead of waiting in real time

11. Assumptions & Dependencies

- The dev branch is deployable at the end of each sprint.
- MongoDB Atlas, Render and Mailtrap free tiers stay available.
- Five classmates are available for the UAT session.
- The SRS requirement IDs stay stable; changes go through a Jira change ticket.

12. Suspension & Resumption Criteria

Suspend testing if:

- Login or the seed script fails, which blocks more than 30% of test cases.
- The test environment is down for more than 4 hours.
- A Critical security defect is found (stop and fix before continuing).

Resume testing if:

- The blocking defect is fixed and its retest passes.
- The environment is back and smoke tests (TC-AUTH-03, TC-SRCH-01) pass.

13. Test Case Management & Traceability

Each test case is a Jira issue linked to the user story and to its SRS IDs. The RTM below maps requirements to test cases; SRS Appendix C extends it with design and code references.

TC-AUTH-01: Register a student and reject a duplicate email

Field	Value
Requirement(s)	AUTH-1, AUTH-2
Type	Functional
Preconditions	No account exists for the test email.
Steps	1. POST /api/auth/register with valid student data. 2. Repeat the same request.
Test data	name=Test Student, email=ts1@pesu.pes.edu, password=Passw0rd1, srn=PES1UG24CS999
Expected result	First call returns 201 with role STUDENT and status ACTIVE. Second call returns 409 with message "An account with this email already exists".
Actual result / Status	Not run

TC-AUTH-02: Reject weak passwords

Field	Value
Requirement(s)	AUTH-3
Type	Functional (negative)
Preconditions	None.
Steps	POST /api/auth/register once with each password in the data column.
Test data	"abc", "abcdefgh", "12345678"
Expected result	Each call returns 400 and names the failed rule; no account is created.
Actual result / Status	Not run

TC-AUTH-03: Login success, then lockout after 5 failures

Field	Value
Requirement(s)	AUTH-4, AUTH-5
Type	Functional
Preconditions	Active student account exists.
Steps	1. Login with the correct password. 2. Login 5 times with a wrong password. 3. Login with the correct password.
Test data	email=ts1@pesu.pes.edu
Expected result	Step 1 returns 200 with a token whose exp is 60 minutes ahead. Step 3 returns 423 with the unlock time.
Actual result / Status	Not run

TC-AUTH-04: Role and ownership checks on protected routes

Field	Value
Requirement(s)	AUTH-6, SEC-3
Type	Security
Preconditions	Tokens for one student, two recruiters (R1 owns listing L1) and one admin.
Steps	1. Student calls POST /api/listings. 2. R2 calls GET /api/listings/L1/applications. 3. R2 downloads the resume of a student who applied only to L1.
Test data	Seeded users and listing L1
Expected result	All three requests return 403 and nothing is changed.

Field	Value
Actual result / Status	Not run

TC-PROF-01: Resume upload type and size limits

Field	Value
Requirement(s)	PROF-2
Type	Functional (boundary)
Preconditions	Logged in as student.
Steps	Upload each file in the data column.
Test data	resume.pdf 1.9 MB, resume.pdf 2.1 MB, resume.docx 100 KB
Expected result	Only the 1.9 MB PDF is accepted (200). The others return 400 with the reason.
Actual result / Status	Not run

TC-LST-01: Create a listing, reject bad dates, auto-close

Field	Value
Requirement(s)	LST-1, LST-2, LST-4
Type	Functional
Preconditions	Approved recruiter logged in.
Steps	1. Create listing with deadline tomorrow. 2. Create listing with deadline yesterday. 3. Set L1 deadline to now+1 min and wait 6 minutes.
Test data	stipend=15000, skills=[React, Node]
Expected result	Step 1 returns 201 OPEN. Step 2 returns 400. After step 3 the listing status is CLOSED.
Actual result / Status	Not run

TC-SRCH-01: Keyword search with filters and paging

Field	Value
Requirement(s)	SRCH-1, SRCH-2, SRCH-3
Type	Functional
Preconditions	5,000 seeded listings, 37 of them match the query.
Steps	GET /api/listings?q=react&mode=remote&minStipend=10000&sort=deadline&page=2
Test data	Seed script data
Expected result	Returns 17 results on page 2, total=37, every result OPEN, remote, stipend >= 10000, sorted by deadline.
Actual result / Status	Not run

TC-APP-01: Apply once and record history

Field	Value
Requirement(s)	APP-1, APP-4
Type	Functional
Preconditions	Student eligible, listing L1 open.
Steps	1. POST /api/applications for L1. 2. Repeat the same request. 3. GET /api/applications/mine.
Test data	listingId=L1, coverNote of 200 chars
Expected result	Step 1 returns 201 APPLIED. Step 2 returns 409 ALREADY_APPLIED. Step 3 shows one application with one history entry.

Field	Value
Actual result / Status	Not run

TC-APP-02: Reject applying to a closed listing or when not eligible

Field	Value
Requirement(s)	APP-2, APP-3
Type	Functional (negative)
Preconditions	L2 is CLOSED. L3 needs CGPA 8.0, student has 7.4.
Steps	Apply to L2, then apply to L3.
Test data	listingIds L2, L3
Expected result	L2 returns 409 LISTING_CLOSED. L3 returns 422 with message naming 8.0 as the required CGPA.
Actual result / Status	Not run

TC-REV-01: Status transitions and notification

Field	Value
Requirement(s)	REV-2, REV-3, REV-4
Type	Functional
Preconditions	Application A1 is APPLIED; recruiter owns its listing.
Steps	1. Move A1 to SHORTLISTED. 2. Move A1 to APPLIED. 3. Move A1 to REJECTED without reason. 4. Check student notifications and mail sandbox.
Test data	A1
Expected result	Step 1 returns 200. Step 2 returns 422 INVALID_TRANSITION. Step 3 returns 400. Step 4 shows one in-app notification and one email for the shortlist.
Actual result / Status	Not run

TC-TST-01: Resume an attempt after refresh with no lost answers

Field	Value
Requirement(s)	TST-3, TST-4, SAFE-1
Type	Functional / Reliability
Preconditions	Test T1 of 20 questions, 30 minutes.
Steps	1. Start T1 and answer 5 questions. 2. Close the browser tab. 3. Reopen and start T1 again.
Test data	T1
Expected result	The same attempt is returned with the 5 answers and the remaining time counted from the original start.
Actual result / Status	Not run

TC-TST-02: Auto-submit at end time and score by topic

Field	Value
Requirement(s)	TST-5, TST-6
Type	Functional
Preconditions	Test T2 with 10 minute duration; student answers 6 of 10 correctly.
Steps	Start T2, answer, then wait past the end time. Send one more answer 40 s after end time.
Test data	T2 answer key

Field	Value
Expected result	Attempt status becomes SUBMITTED with score 6/10 and per-topic percentages. The late answer returns 409 ATTEMPT_CLOSED.
Actual result / Status	Not run

TC-ANL-01: Placement statistics and CSV export

Field	Value
Requirement(s)	ANL-2, ANL-3
Type	Functional
Preconditions	Seed: 40 students in CSE 2027, 12 with an offer.
Steps	Admin opens statistics filtered by CSE and 2027, then exports CSV.
Test data	Seed data
Expected result	Dashboard shows 12 of 40 placed (30%). CSV has 40 rows with the same numbers.
Actual result / Status	Not run

TC-PERF-01: Search response time under load

Field	Value
Requirement(s)	PERF-1, PERF-2
Type	Performance
Preconditions	5,000 listings seeded, server on test VM.
Steps	Run JMeter plan: 50 virtual users, 5 minutes, mixed search queries.
Test data	search_load.jmx
Expected result	95th percentile under 2 s for search and under 1 s for other calls; error rate under 1%.
Actual result / Status	Not run

TC-SEC-01: Password storage and NoSQL injection

Field	Value
Requirement(s)	SEC-1, SEC-4
Type	Security
Preconditions	None.
Steps	1. Register a user and read the stored document from MongoDB. 2. POST /api/auth/login with body {"email": {"$ne": null}, "password": {"$ne": null}}.
Test data	-
Expected result	Stored password is a bcrypt hash starting with $2b$10 or higher. Step 2 returns 400 and no token.
Actual result / Status	Not run

TC-SEC-02: Rate limit and HTTPS only

Field	Value
Requirement(s)	SEC-5, SEC-2
Type	Security
Preconditions	Deployed build.
Steps	1. Send 11 login requests in one minute from one IP. 2. Request the site over plain HTTP.
Test data	-

Field	Value
Expected result	The 11th request returns 429. Plain HTTP is redirected to HTTPS; TLS version reported is 1.2 or higher.
Actual result / Status	Not run

Requirements Traceability Matrix:

Requirement ID	Test case(s)
AUTH-1	TC-AUTH-01
AUTH-2	TC-AUTH-01
AUTH-3	TC-AUTH-02
AUTH-4	TC-AUTH-03
AUTH-5	TC-AUTH-03
AUTH-6	TC-AUTH-04
PROF-1	Unit/integration suite only
PROF-2	TC-PROF-01
PROF-3	Unit/integration suite only
PROF-4	Unit/integration suite only
LST-1	TC-LST-01
LST-2	TC-LST-01
LST-3	Unit/integration suite only
LST-4	TC-LST-01
SRCH-1	TC-SRCH-01
SRCH-2	TC-SRCH-01
SRCH-3	TC-SRCH-01
SRCH-4	Unit/integration suite only
SRCH-5	Unit/integration suite only
APP-1	TC-APP-01
APP-2	TC-APP-02
APP-3	TC-APP-02
APP-4	TC-APP-01
APP-5	Unit/integration suite only
REV-1	Unit/integration suite only
REV-2	TC-REV-01
REV-3	TC-REV-01
REV-4	TC-REV-01
TST-1	Unit/integration suite only
TST-2	Unit/integration suite only
TST-3	TC-TST-01
TST-4	TC-TST-01
TST-5	TC-TST-02
TST-6	TC-TST-02
ANL-1	Unit/integration suite only
ANL-2	TC-ANL-01
ANL-3	TC-ANL-01

Requirement ID	Test case(s)
PERF-1	TC-PERF-01
PERF-2	TC-PERF-01
PERF-3	Reviewed in Sprint 3
SAFE-1	TC-TST-01
SAFE-2	Reviewed in Sprint 3
SEC-1	TC-SEC-01
SEC-2	TC-SEC-02
SEC-3	TC-AUTH-04
SEC-4	TC-SEC-01
SEC-5	TC-SEC-02
SEC-6	Security checklist 5.1

Total: 16 test cases, 12 functional and 4 non-functional (performance and security).

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed and % passed/failed per sprint
- Defects by severity and by module; defect density per 1,000 lines
- Defect age (days open)
- Requirement coverage: share of REQ IDs with a passed test
- Server statement coverage from Jest

Reports:

- Sprint test status on the Jira dashboard, reviewed at each sprint review
- CI coverage report on every PR
- Final test summary report on 23-Nov-2026

15. Approvals

Role	Name	Signature / Date
QA Lead	Guru R Bharadwaj	
Dev Lead	E S Sukruth	
Product Owner		